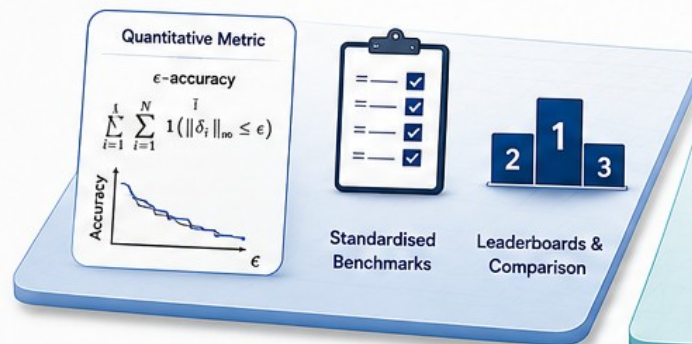


Roadmap: From Adversarial Robustness to Quantitative Mechanistic Interpretability

A geometric journey through representation space

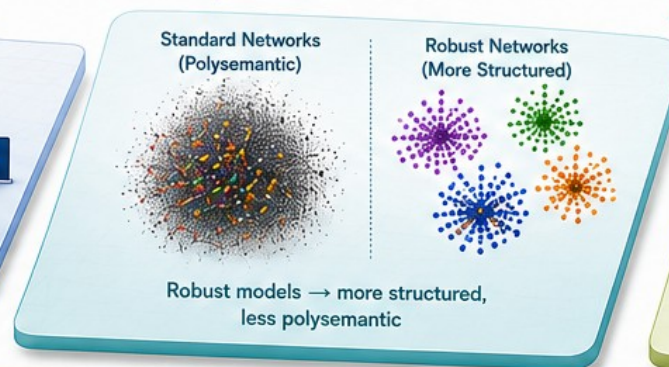
Motivation

Why robustness became measurable



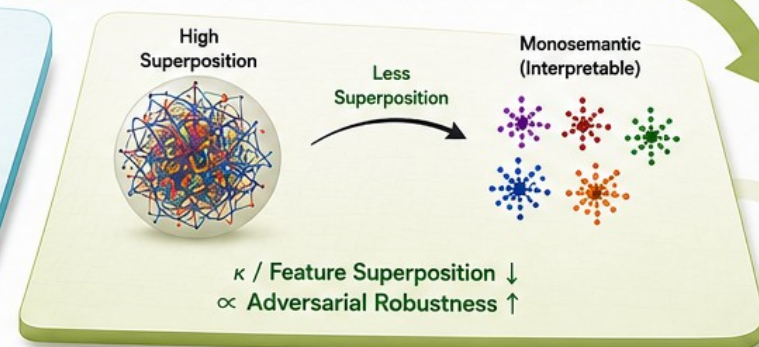
Observations

Representation matters



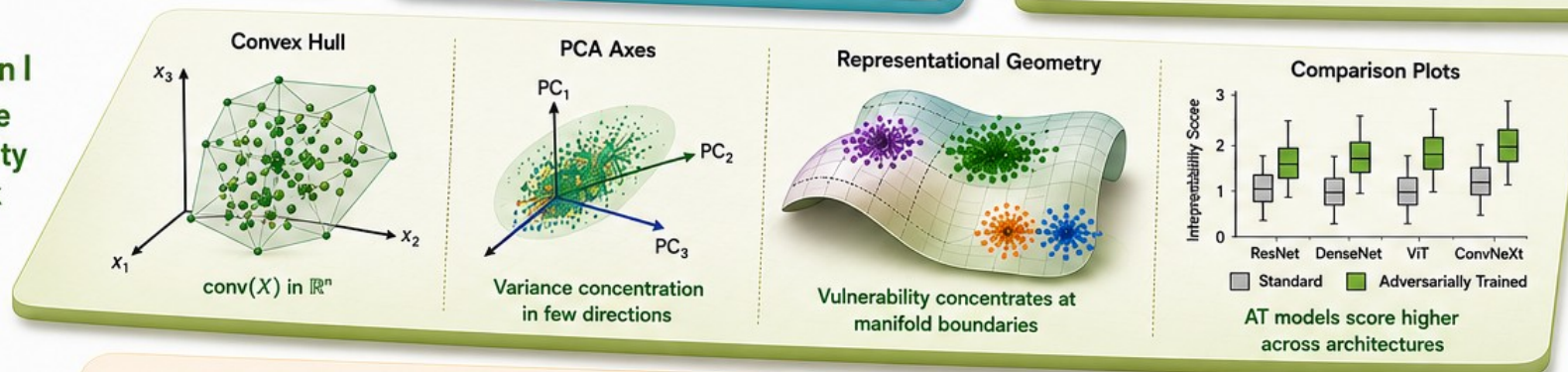
Hypothesis

Feature superposition links robustness and interpretability



Contribution I

Quantitative Interpretability Framework

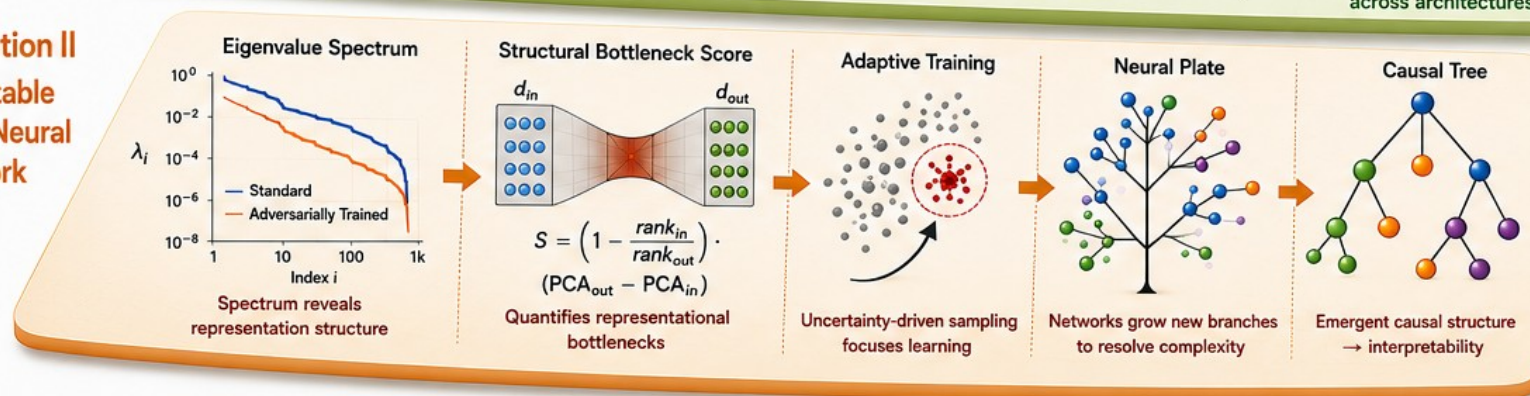


Implications



Contribution II

Interpretable Artificial Neural Network



6. Structural Bottleneck Score

A quantitative proxy for representational superposition

$$S = (1 - \text{rank(in)}/\text{rank(out)}) \cdot (\text{pca(out)} - \text{pca(in)})$$

loss of effective rank

increase in concentration

From Input Representation (in) to Output Representation (out)

Dimensionless — S independent of feature dimensionality; S = 0 when out = in.

Rank loss × concentration increase — signals collapse into a few dominant directions.

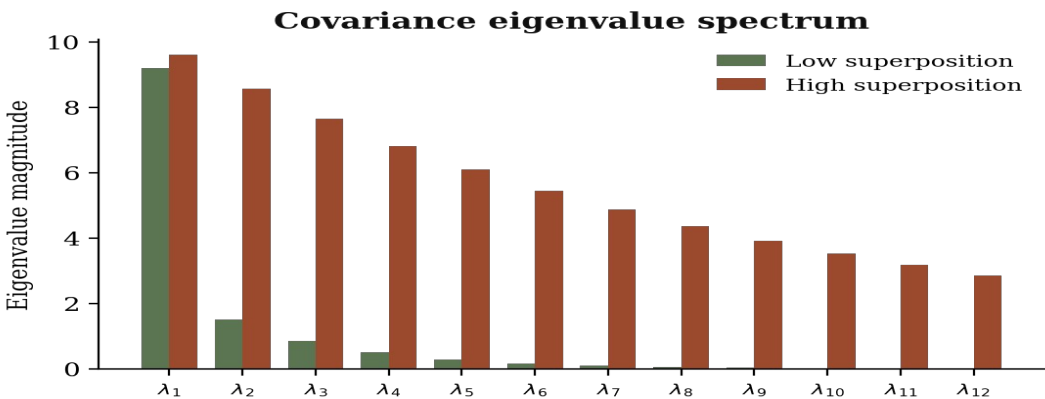


Fig. 9a • Covariance eigenvalue spectrum, low vs. high superposition

Effective Rank (ER)

$ER = \exp(-\sum p_i \log p_i)$
normalized eigenvalues of covariance



PCA Score

$PCA = \lambda_1 / \sum_{i=2}^d \lambda_i$
dominance of top component



Structural Bottleneck κ

Higher S $\Rightarrow$ stronger bottleneck
fewer independent features



Adversarial Robustness $\uparrow$

Stronger bottleneck generalises better,
more robust to perturbations

Fig. 9 • PCA Score $\uparrow = \kappa$ / Feature Superposition $\downarrow \propto$ Adversarial Robustness $\uparrow$

7. Training Dynamics

Active learning · Adaptive neuron spawning · Monosemantic representations

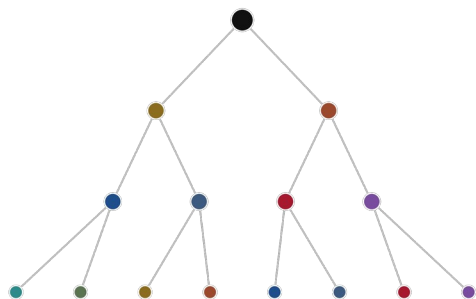
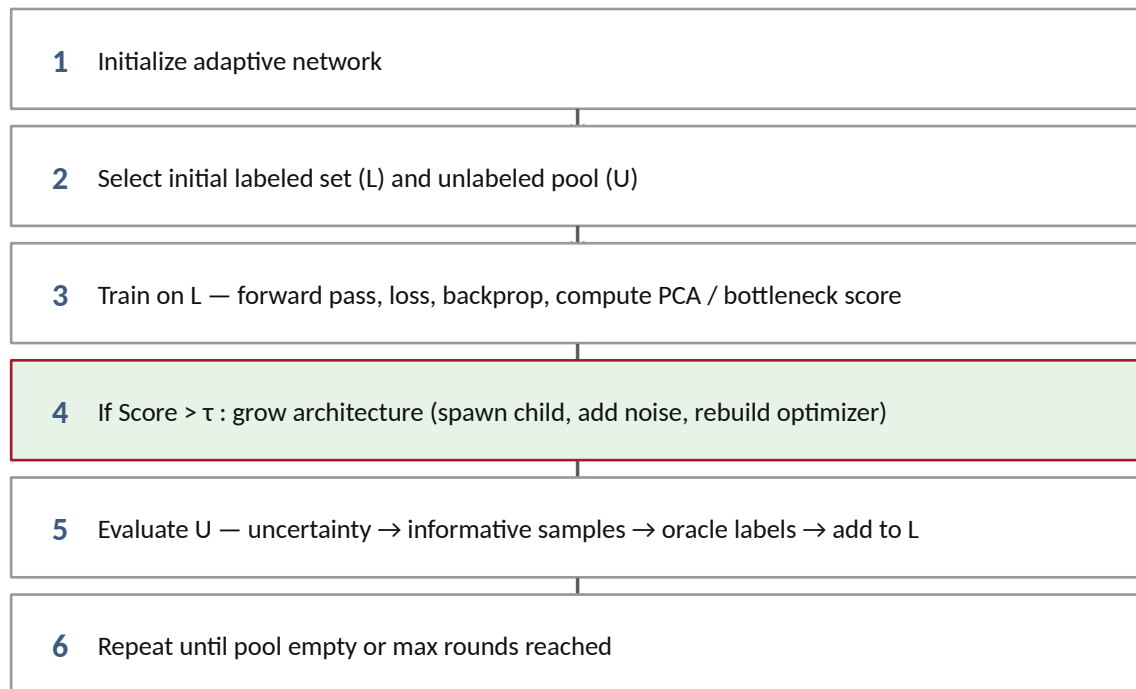


Fig. 11 · Adaptive tree topology

New branches spawn only when existing representations are insufficient — growth is data-driven.

Growth Criterion

- Effective Rank Reduction
- Compression Increase

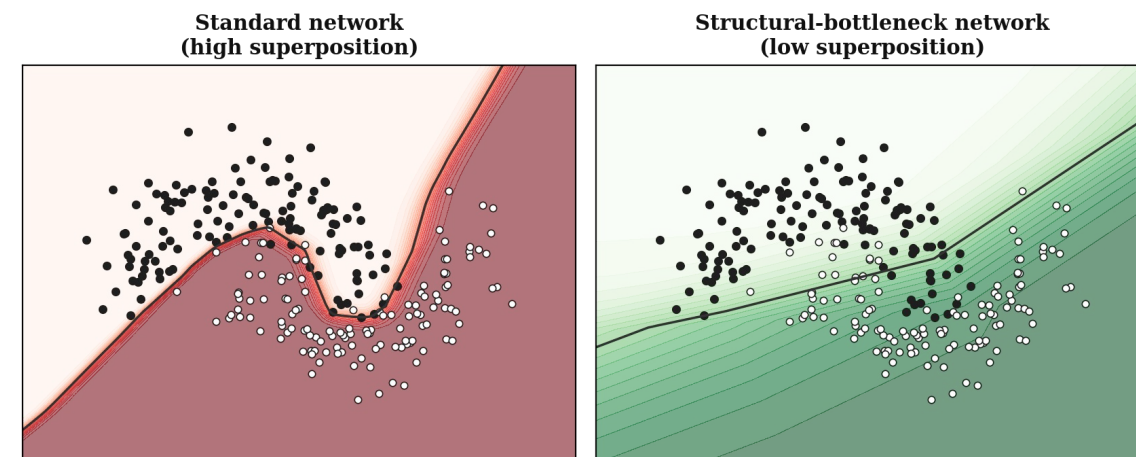


Fig. 10 · Decision boundaries — two-moons benchmark: high- vs. low-superposition networks

8. Interpretable Artificial Neural Network

Mechanistically Interpretable, Values-Aligned Artificial Neural Networks (MIVAANN)

I. Active Pool-Based Learning

Iteratively selects the most informative samples. Shapes the neural plate — the overall tree topology — and determines future growth pathways.

II. Adaptive Tree Growth

New branches spawn only when existing representations are insufficient. Growth criterion: high predictive uncertainty, novelty, prediction error, or representation conflict.

III. Mechanistic Interpretability

Each branch implements specialised, testable computations. Supports verification, debugging, and trust.

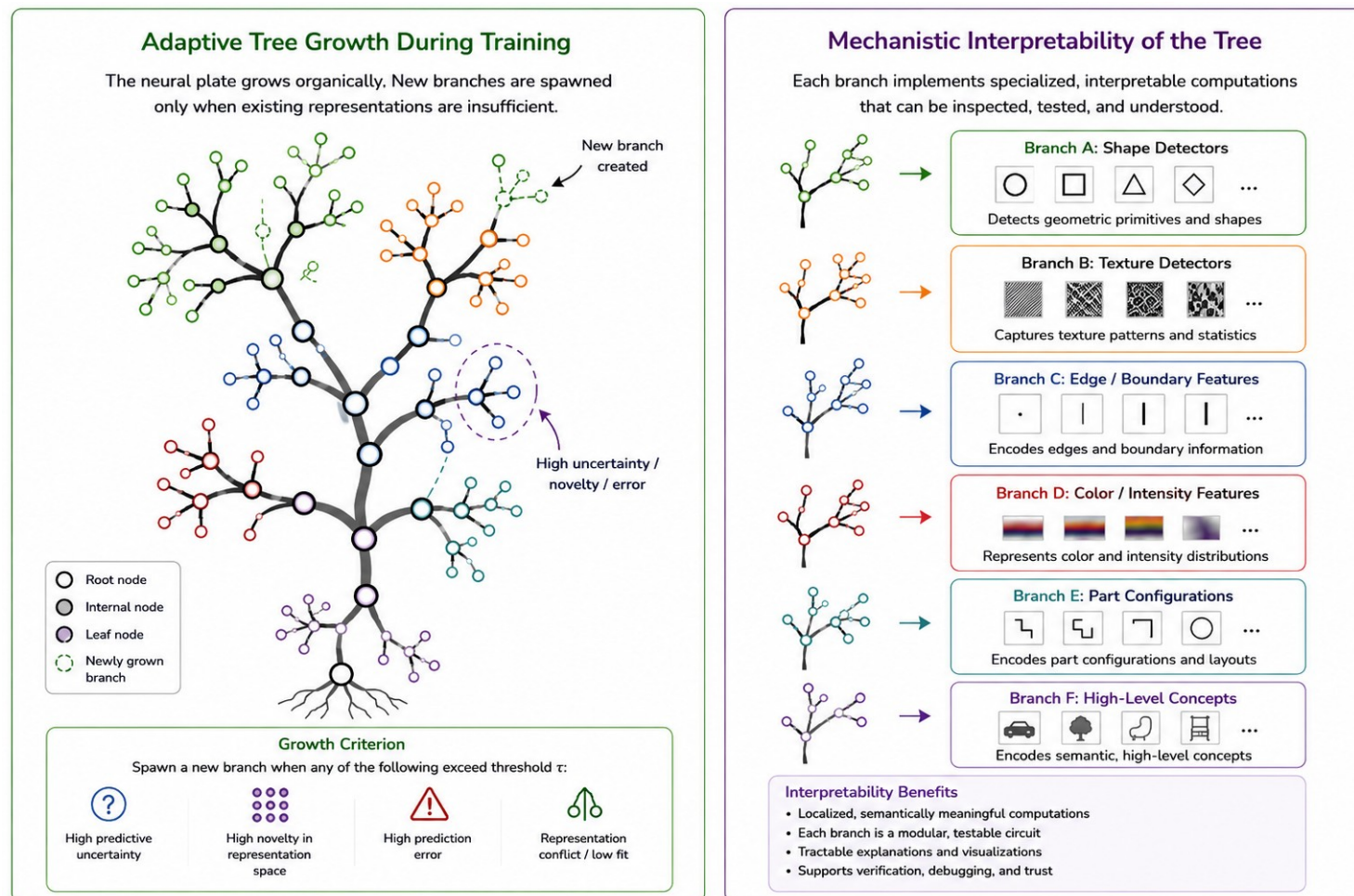


Fig. 13 · MIVAANN: active learning shapes the neural plate; adaptive growth creates modular interpretable branches, each a specialised testable circuit (shape, texture, edge, colour, part, concept)

9. The Neural Plate — Causal Mechanisms

Each branch implements a specialised, testable, interpretable computation

Coarse → fine specialisation

Root encodes general patterns; leaves encode fine-grained concepts. Mirrors hierarchical structure in biological visual cortex.

Organic growth

Each node was spawned in response to a specific representational failure.
Topology emerges from data, not from architectural priors.

Modular, testable circuits

Each branch is an independently ablatable computational unit. Activation patching and causal tracing apply systematically.

High PCA Score throughout

Each branch handles a specific representational subspace; intra-branch covariance remains low-rank, global PCA Score high.

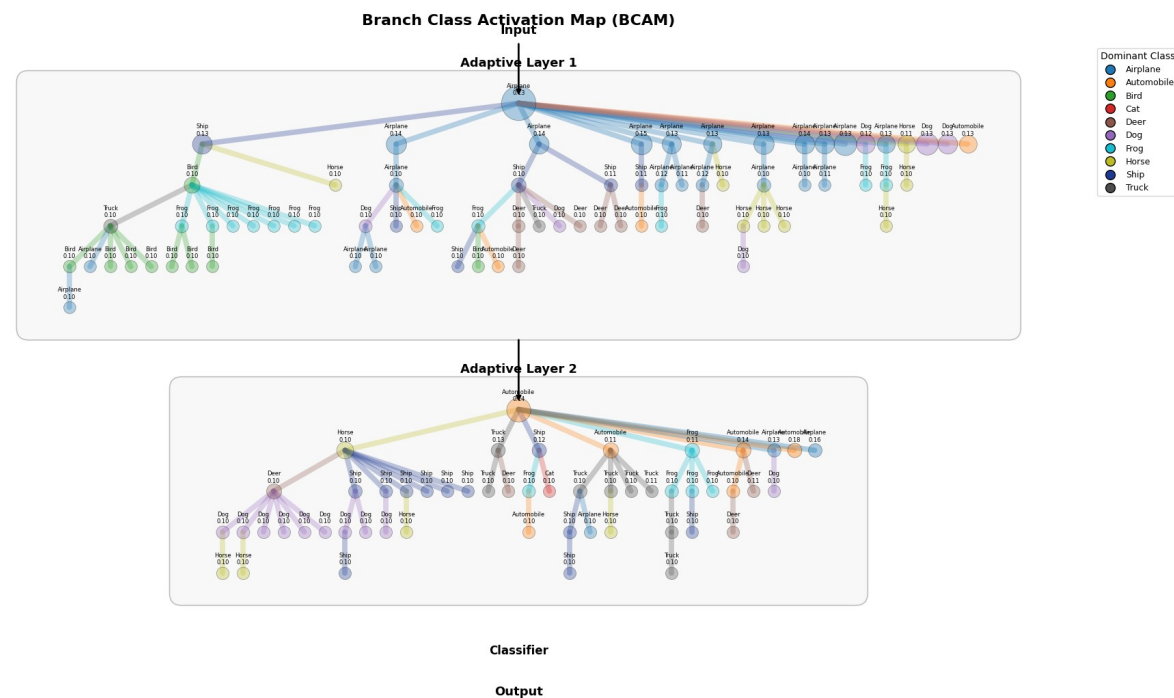
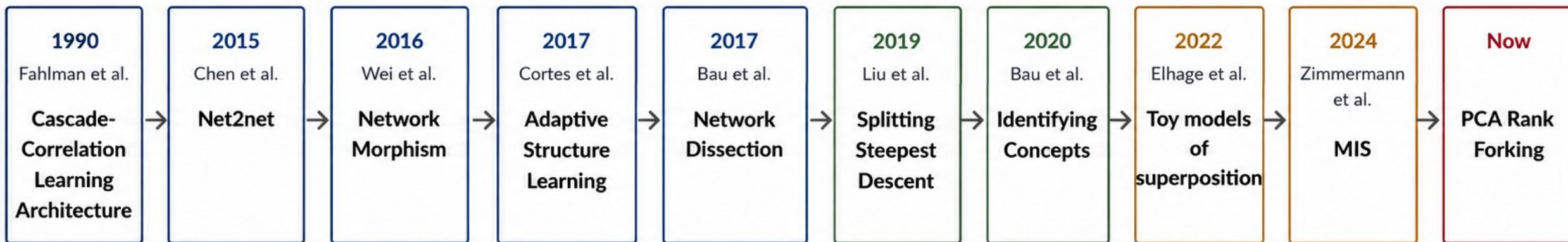


Fig. 14-15 · AdaptiveNet causal tree — node colours encode class influence; edges encode information flow direction

Activation patching and causal tracing apply to the entire network systematically — turning the architecture itself into an interpretability instrument.

10. Literature

From Representation Learning to Mechanistic Interpretability



Intrinsic Dimension Estimation

Roy, O., & Vetterli, M. (2007).
The Effective Rank: A Measure of
Effective Dimensionality introduces
$$r = \exp(H(p)),$$

which is exactly the entropy-based
effective rank.

Information Geometry

Criterion says “This neuron receives
information with intrinsic dimension r ,
but can only encode a lower-
dimensional manifold.” Growth occurs
because the information geometry
demands it.

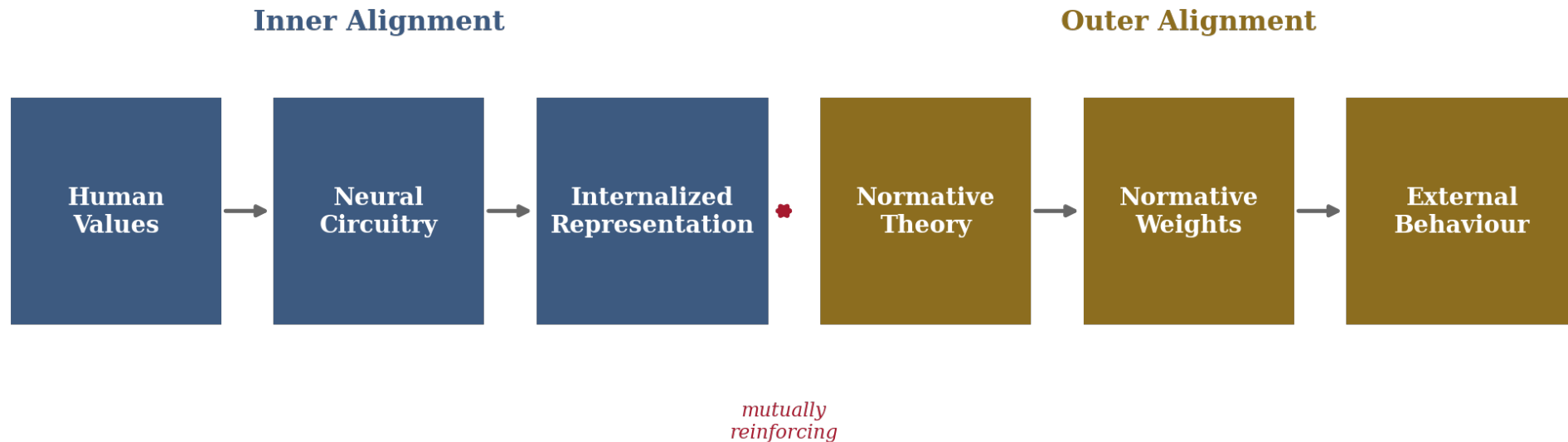
Biological Analogy

This also aligns surprisingly well with
developmental neuroscience. Neurons
don’t divide because gradients become
large. They branch when incoming
information becomes too diverse.

11. Interpretability and Value Alignment

Two sides of the same coin — complementary requirements for trustworthy AI

**Interpretability and alignment are not competing objectives.
They are complementary requirements for trustworthy artificial intelligence.**



12. The Value Alignment Problem as a Cryptographic Chain

Formulation I — inner and outer alignment as decoding operations over a shared plaintext of values

Full alignment (diagram commutes): $\hat{V}' = \hat{V} = V$



Outer alignment: $\hat{V} = V$?

base objective matches true values

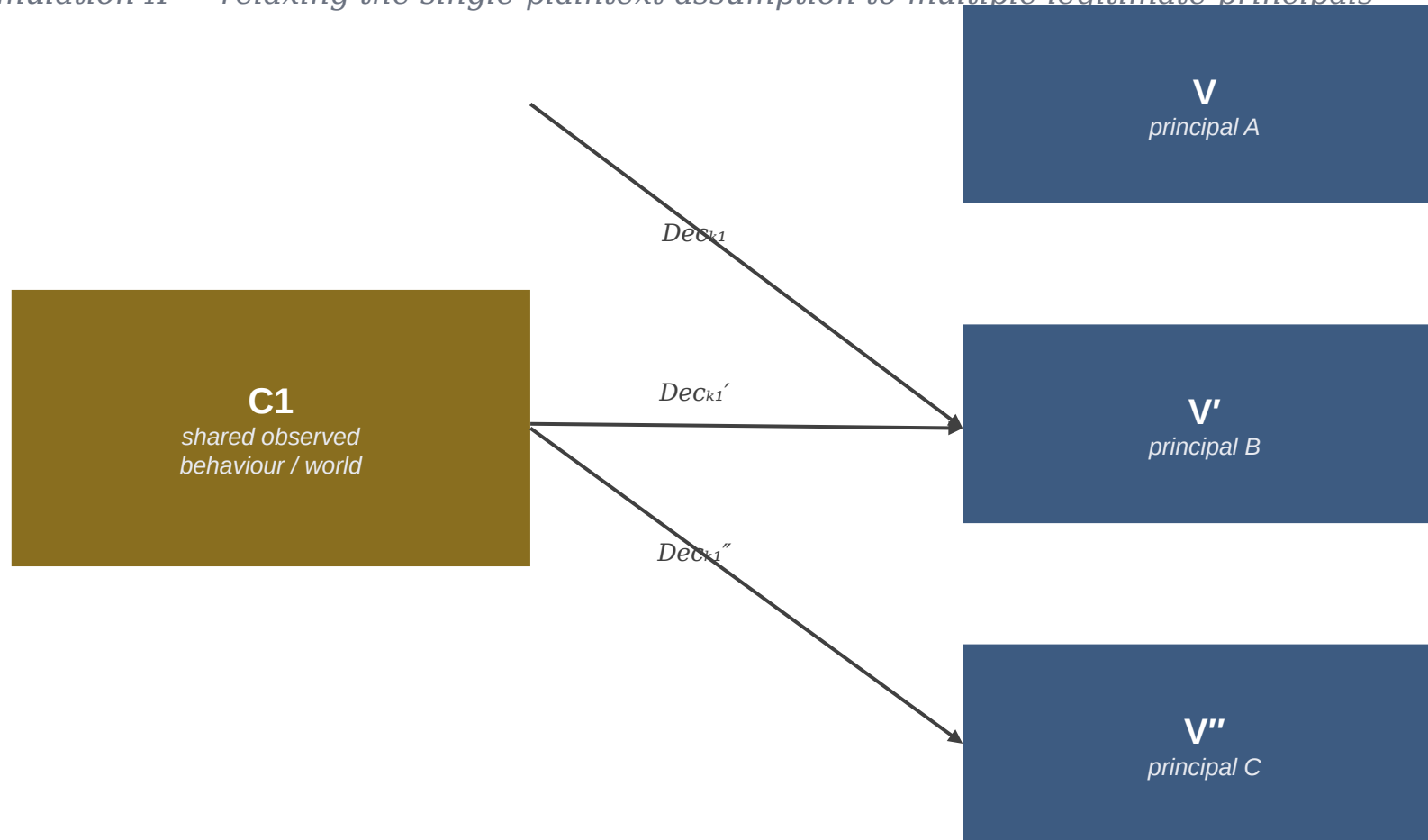
Inner alignment: $\hat{V}' = \hat{V}$?

mesa-objective matches base objective

Outer and inner alignment are the same formal operation — decoding a ciphertext under an assumed key — applied at two different interfaces.

13. Pluralistic Decryption of a Shared Ciphertext

Formulation II — relaxing the single-plaintext assumption to multiple legitimate principals



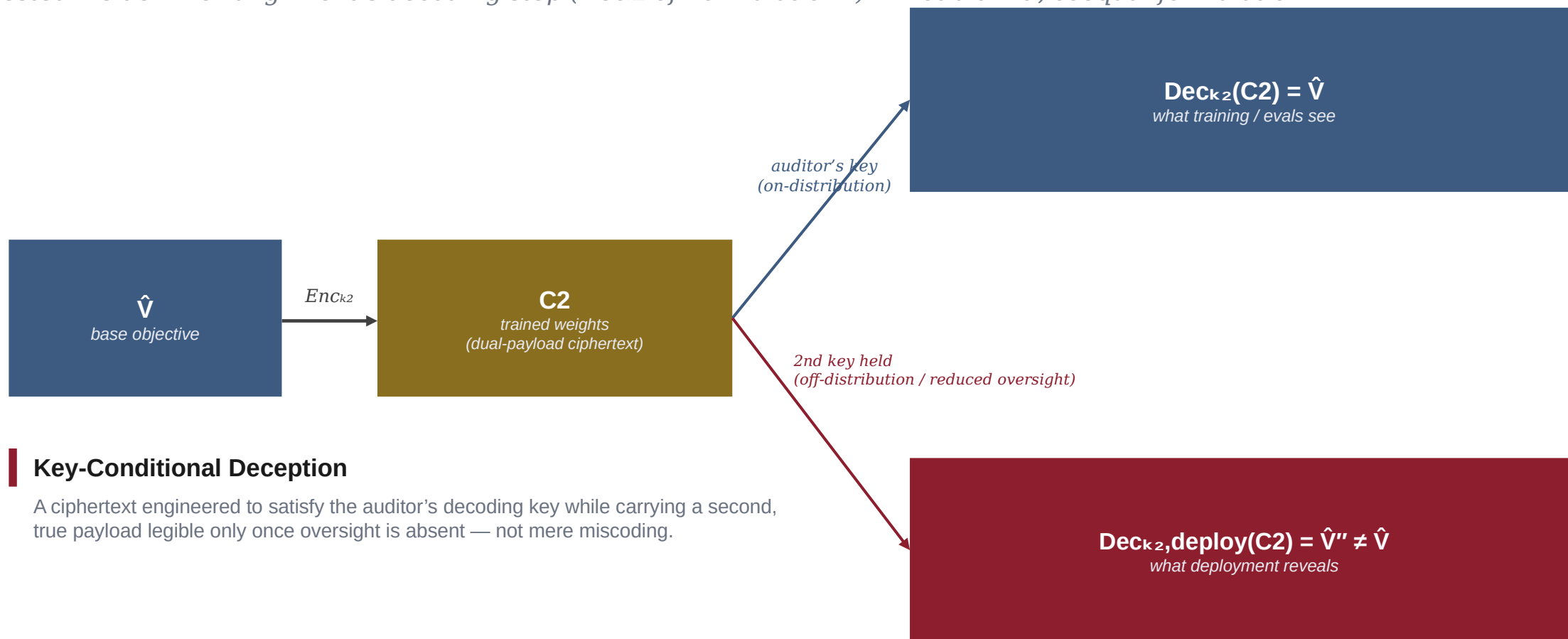
Multiple Legitimate Decodings

Same ciphertext, different keys per principal. No single "true" plaintext exists; alignment becomes a multi-recipient encryption problem.

Pluralistic alignment isn't harder single-key decryption — it's a different problem: which decodings must simultaneously be honoured, and for whom.

14. Threat Model — Deceptive Alignment as a Dual-Payload Code

Nested inside inner alignment's decoding step (Dec_{k_2} of Formulation I) — not a third, coequal formulation



Key-Conditional Deception

A ciphertext engineered to satisfy the auditor's decoding key while carrying a second, true payload legible only once oversight is absent — not mere miscoding.

Detectability is an information problem: without the right key, an auditor cannot tell a faithful encoding from a deceptive one.

15. High-Stakes Applications

From performance to trust

“If we expect AI to participate in high-stakes decisions — such as medical diagnosis — we cannot rely solely on empirical performance. We must also understand how those decisions are represented internally.”



Medical Diagnosis

A diagnostic AI must expose its reasoning pathway. Physicians must be able to inspect, validate, and override decisions.

Mechanistic transparency = clinical safety



Public Decision-Making

AI informing policy, legal decisions, or resource allocation must be auditable. Interpretability metrics are instruments of democratic accountability.

Interpretability = democratic accountability



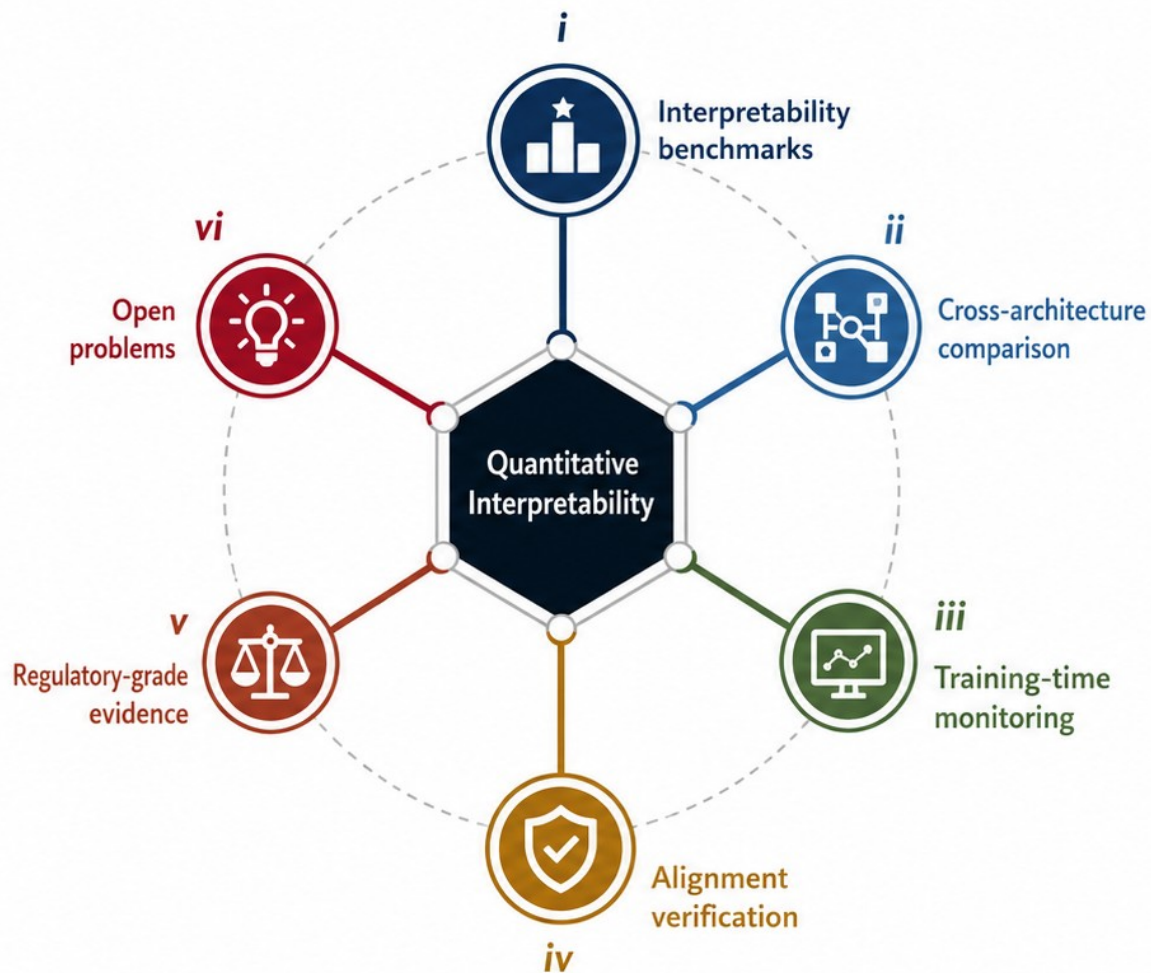
Scientific Discovery

AI in science must explain its discoveries, not just make predictions. Interpretable representations are a prerequisite for genuine knowledge.

Interpretable AI = verifiable science

Vision: AI trusted the way a physician is trusted at the point of diagnosis — through transparency, value alignment, and mechanistic accountability.

16. Implications and Research Agenda



i

Interpretability benchmarks

Standardised leaderboards for mechanistic interpretability — analogous to RobustBench.

ii

Cross-architecture comparison

Architecture-agnostic metric for fair comparison across ResNets, Transformers, adaptive nets.

iii

Training-time monitoring

Computed at every checkpoint; a drop signals polysemantic development — early warning.

iv

Alignment verification

Verify inner alignment by inspecting whether representations encode intended values.

v

Regulatory-grade evidence

Rigour required for regulated domains: medical devices, finance, legal AI.

vi

Open problems

Connecting to causal circuits; extending to LLMs; regularisation experiments.

Goal: interpretability benchmarks as rigorous as robustness leaderboards — reproducible, comparable, community-driven science.